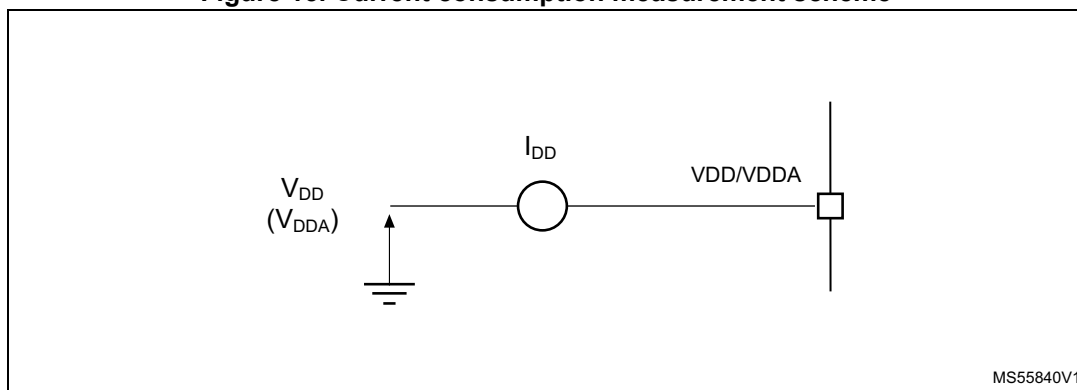


5.1.7 Current consumption measurement

Figure 15. Current consumption measurement scheme



5.2 Absolute maximum ratings

Stresses above the absolute maximum ratings listed in [Table 21](#), [Table 22](#) and [Table 23](#) may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability. The device mission profile (application conditions) is compliant with the JEDEC JESD47 qualification standard.

All voltages are defined with respect to V_{SS} .

Table 21. Voltage characteristics

Symbol	Ratings	Min	Max	Unit
V_{DD}	External supply voltage	- 0.3	4.0	V
V_{DDIO2}	External supply voltage for selected I/Os	- 0.3	4.0	V
V_{REF+}	External voltage on VREF+ pin	- 0.3	$\text{Min}(V_{DD} + 0.4, 4.0)$	V
$V_{IN}^{(1)}$	Input voltage on pin	- 0.3	$V_{DDIOx} + 4.0^{(2)(3)}$	V

1. V_{IN} maximum must always be respected. Refer to [Table 22](#) for the maximum allowed injected current values.
2. To sustain a voltage higher than 4 V the internal pull-up/pull-down resistors must be disabled.
3. When an FT_a pin is used by an analog peripheral such as ADC, the maximum V_{IN} is 4 V.

Table 22. Current characteristics

Symbol	Ratings	Max	Unit
$I_{VDD/VDDA, VDDIO2}$	Current into VDD/VDDA and VDDIO2 power pins (source)	100	mA
$I_{VSS/VSSA, VSS}$	Current out of VSS/VSSA and VSS ground pins (sink)	100	mA
$I_{IO(PIN)}$	Output current sunk by any I/O and control pin	20	mA
	Output current sourced by any I/O and control pin	20	
$\Sigma I_{(PIN)}$	Total output current sunk by sum of all I/Os and control pins ⁽¹⁾	80	mA
	Total output current sourced by sum of all I/Os and control pins ⁽¹⁾	80	